

The E0 pair channel in ${}^3\text{He}(n, \cdot){}^4\text{He}$

A second, γ -dark way to make e^+e^- pairs — why it matters, where it lives in energy, and how big it might be

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Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff’s direction; physics, numbers and text reviewed by D. Neff.

Part I — Summary of findings

There are two distinct ways a captured neutron makes an e^+e^- pair in ${}^4\text{He}$, and they are easy to confuse. They are *not* two decays of one state — they are **different capture channels**, picked out by how the neutron and ${}^3\text{He}$ spins line up:

- **M1 channel** (${}^3S_1 \rightarrow 0^+$): emits a real γ ; only $\sim 0.35\%$ of those γ ’s convert to a pair. This is the ordinary radiative capture (the $54\mu\text{b}$). It *grows* toward MeV, so its pairs pile up in the **MeV** region.
- **E0 channel** (${}^1S_0 \rightarrow 0^+$): a $0^+ \rightarrow 0^+$ transition, for which a γ is *forbidden* — so it can *only* make a pair, **100%** of the time. It is s-wave and boosted by the 20.21 MeV 0^+ state just below threshold, so its pairs pile up at **thermal / sub-keV**.

Key consequences: (i) a $0^+ \rightarrow 0^+$ transition can go *only* by E0 — not M1 (the same $0 \rightarrow 0$ rule forbids both the photon and M1). (ii) The E0 channel is **missing from the standard tools** (ENDF, Geant4) *and* from our generator — it is γ -dark, so nothing counts it. (iii) Its energy dependence is the *opposite* of the M1/E1 pairs. (iv) **Its absolute size is not yet known.** A literature search found the ${}^4\text{He}$ 0^+ E0 strength is real and measured, but the capture rate has never been computed; a transparent estimate brackets it at $f = \sigma_{\text{E0}}/\sigma_{\text{M1}} \sim 10^{-3}\text{--}10^{-2}$, i.e. **E0 pairs $\sim 0.3\text{--}3\times$ the ordinary M1 pairs at low neutron energy** — exactly Alberto’s “same order of magnitude” intuition. Pinning the number needs a ${}^4\text{He}$ R-matrix calculation.

What this note is. Part II explains the nuclear physics from the ground up, in plain terms, with diagrams. Part III is the magnitude: what we can and cannot say, the literature dig, and the bracket. The headline picture is Fig. 4: **E0 pairs at low E_n , M1/E1 pairs at MeV.**

1 What happens when a neutron is captured by ${}^3\text{He}$

A slow neutron and a ${}^3\text{He}$ fuse into an excited ${}^4\text{He}^*$ nucleus (${}^4\text{He}^*$, $\approx 20.6\text{ MeV}$). That excited nucleus almost always just breaks back apart into $p + {}^3\text{H}$ (the (n,p)t reaction, $\sim 100\%$ — and no pair). Very rarely ($10^{-8}\text{--}10^{-4}$ of the time) it instead drops to the ${}^4\text{He}$ ground state by emitting electromagnetic radiation — **and that rare drop is the only source of e^+e^- pairs**. Figure 1 is the whole logic on one page.

What makes the e^+e^- pairs: E0 and M1 are different capture channels, not two decays of one state

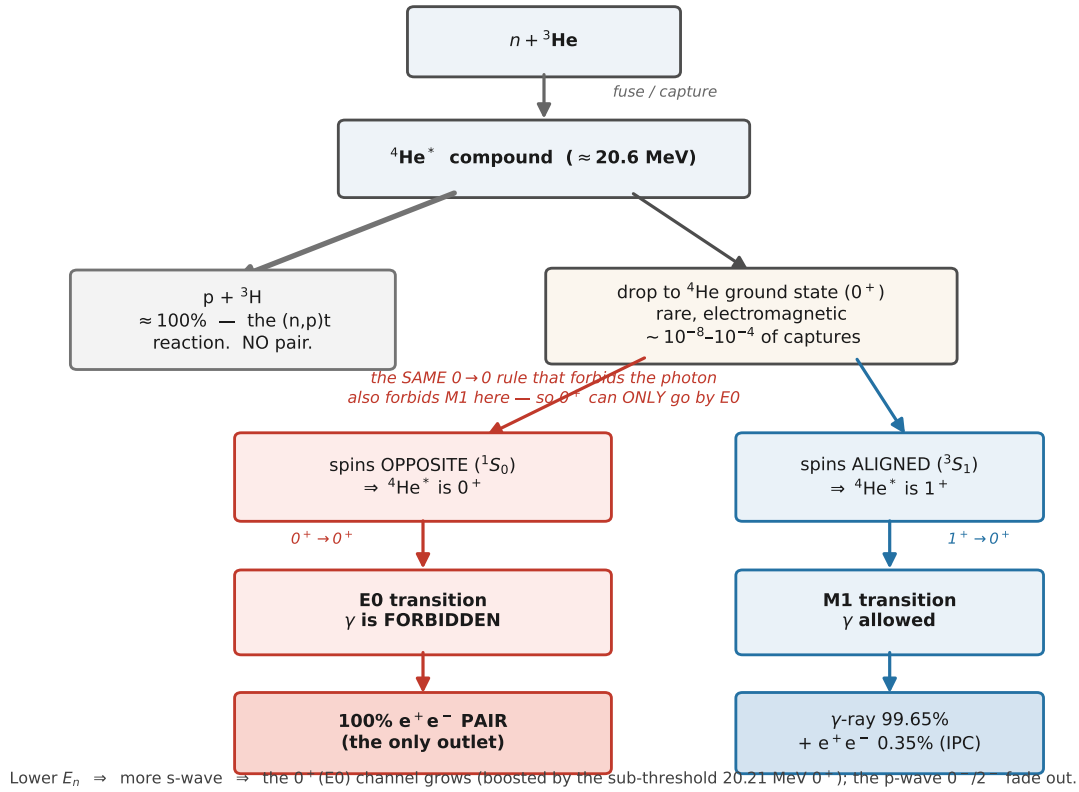


Figure 1: What makes the pairs. The neutron+ ^3He fuse; the result almost always breaks up to $p + t$. The rare electromagnetic drop to the ground state comes in *two flavours*, set by how the two spins line up — and those two flavours are the E0 and M1 channels.

2 The selection rule, in plain words

The ^4He ground state has spin/parity 0^+ . The transition that makes a pair goes from $^4\text{He}^*$ down to that 0^+ ground state. **What kind of radiation is allowed is fixed entirely by the spin and parity of $^4\text{He}^*$** — it is not a free choice.

The crucial fact: *a photon, and an M1, each carry at least one unit of angular momentum.* A transition between two spin-0 states ($0 \rightarrow 0$) has **zero** angular momentum to give away. Therefore, for a $0^+ \rightarrow 0^+$ transition:

- **γ -ray emission: forbidden** (there is no “monopole photon”).
- **M1: forbidden** — same reason, it needs ≥ 1 unit.
- **E0: allowed.** E0 is the loophole: the nucleus hands its energy straight to an e^+e^- pair through a *virtual* photon, which is allowed to be monopole. For a $0^+ \rightarrow 0^+$ transition the pair is the *only* outlet.

So “the 0^+ goes by E0” is forced — it cannot go by M1. This is the point that is easy to trip over: **the M1 in our problem is a different transition entirely**, not an alternative way for the 0^+ to decay.

3 The two channels: E0 vs M1

When the neutron (spin $\frac{1}{2}$) and ^3He (spin $\frac{1}{2}$) capture in s-wave, their spins can be *opposite* or *aligned*, and that decides which $^4\text{He}^*$ state you build:

spins	channel	${}^4\text{He}^*$ is	$\rightarrow$ ground state
opposite	1S_0	0^+	E0: γ forbidden $\Rightarrow$ 100 % pair (γ -dark)
aligned	3S_1	1^+	M1: γ allowed $\Rightarrow \gamma$ (99.65 %) + 0.35 % pair

These are **two parallel doorways**. The all-important practical consequence: the famous radiative-capture number ($54\mu\text{b}$ at thermal, ENDF MT=102) is *only the M1 (1^+) channel* — it counts γ -emitting captures. The E0 (0^+) channel emits no γ , so **it is invisible to that number, to ENDF, and to Geant4**, and it is not in our event generator either. It is missing in three places at once.

The M1 channel’s pairs are the 0.35 % internal-pair-conversion (IPC) tail of its γ : at $\sim 20\text{ MeV}$ the IPC coefficient is $\alpha_{\text{IPC}} \approx 3.5 \times 10^{-3}$ (anchored to the *measured* ${}^{12}\text{C}$ 15.1 MeV M1 value $(3.3 \pm 0.5) \times 10^{-3}$ and the ${}^8\text{Be}$ 18.15 MeV line; BrIcc’s tables stop at 6 MeV so they cannot be used at our energy — see Part III).

4 Why angular momentum sets the energy dependence

Now the energy dependence, which is where it gets interesting. To build a given state you need a specific orbital angular momentum ℓ , because the parity of the $n + {}^3\text{He}$ system is $(-1)^\ell$:

- 0^+ (parity $+$) needs ℓ even $\Rightarrow$ **s-wave** ($\ell = 0$).
- 0^- (parity $-$) needs ℓ odd $\Rightarrow$ **p-wave** ($\ell = 1$).

An s-wave neutron has *no centrifugal barrier* — it gets in for free, and the formation rate behaves as $\sim 1/v$, **growing without limit toward low energy**. A p-wave neutron must climb the barrier, so its formation *vanishes* at threshold and only switches on with energy. Figure 2 makes this concrete with a Breit–Wigner $\times$ penetrability calculation using the measured ${}^4\text{He}$ resonance parameters.

The 0^- question, settled. A natural worry is whether the 0^- state (21.0 MeV) is “energetically forbidden” at low energy. It is not: the $n + {}^3\text{He}$ channel is open at all $E_n > 0$. What suppresses the 0^- below $\sim 0.5\text{ MeV}$ is the *p-wave barrier*, not energetics — and it actually peaks just below 1 MeV. (As a bonus, the 0^- is a *spectator* for the pair signal anyway: $0^- \rightarrow 0^+$ is forbidden for γ , M1 *and* E0, so it simply decays back by particle emission.) **The upshot: as the neutron energy drops, we make relatively more of the 0^+ (E0) doorway.**

5 The cross-section backdrop

For context, Fig. 3 shows the two known ${}^3\text{He}+n$ cross sections. The (n,p)t dominates utterly ($1/v$, 5333 b at thermal). The radiative $\sigma_{n\gamma}$ (the M1/E1, $54\mu\text{b}$ at thermal) is $1/v$ at low energy then *rises* toward the MeV region as the E1 / giant-dipole strength turns on — the ratio $\sigma_{n\gamma}/\sigma_{np}$ climbs $\sim 10^4\times$ from thermal to 10 MeV. That rise is *why the ordinary M1/E1 pairs are a MeV phenomenon*. Note this is the *lumped* radiative number; it contains no E0 (which is γ -dark) and cannot be sliced into multipoles.

6 Where the pairs come from: the punchline

Putting it together: the two pair channels live at **opposite ends** of the energy range (Fig. 4).

- **M1/E1 (blue):** rides the rising $\sigma_{n\gamma}/\sigma_{np}$, so it **peaks in the MeV region**. This is the piece we can already compute from the simulated capture rate $\times \alpha_{\text{IPC}}$ (≈ 2200 pairs/day, concentrated 0.1–10 MeV).

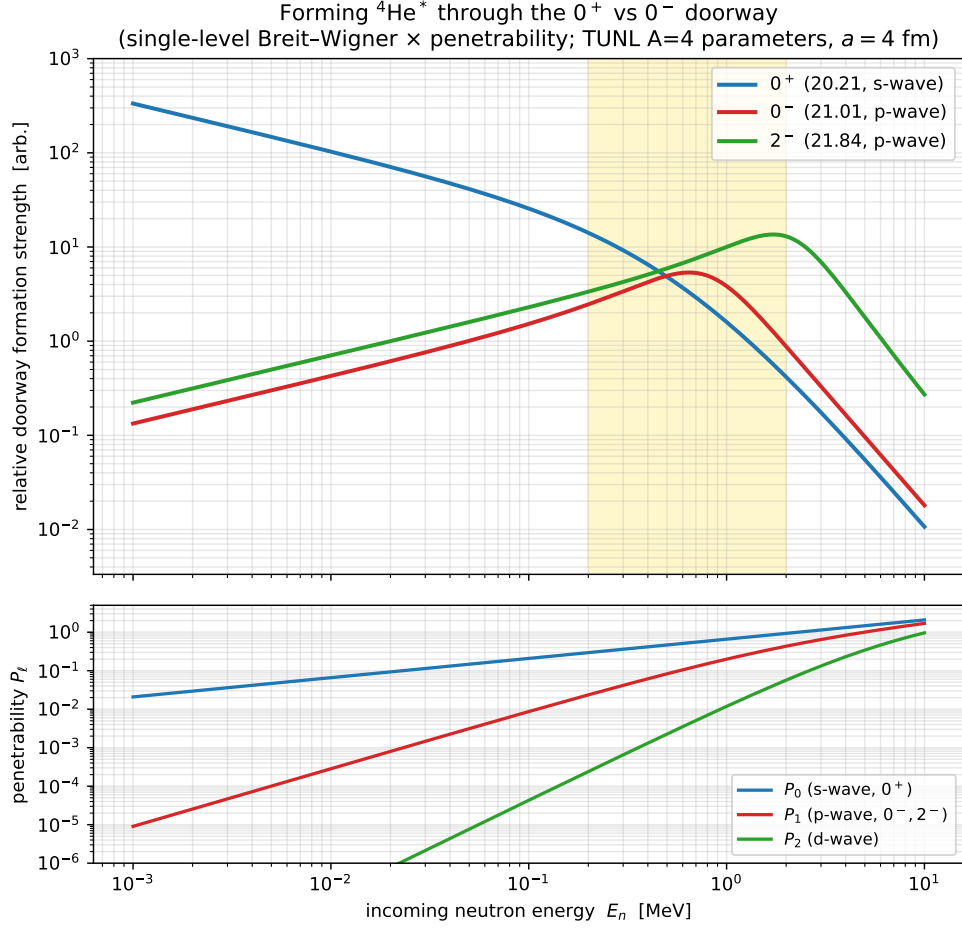


Figure 2: Relative strength for forming the 0^+ (s-wave) vs 0^- (p-wave) doorway, and the underlying penetrabilities (bottom). The 0^+ dominates everywhere below ~ 0.5 MeV — by $\sim 2500\times$ at 1 keV. The 0^- is *not* forbidden below 1 MeV; it is barrier-suppressed and in fact *peaks* at $E_n \approx 0.64$ MeV. (Single-level illustration; absolute vertical offsets need the full R-matrix.)

- **E0 (red):** pure s-wave, boosted by the sub-threshold 0^+ , and with *no* 0.35% penalty (the γ is forbidden, so every such capture is a pair) — so it **piles up at thermal / sub-keV** and fades upward.

The striking implication: **the sub-keV region we had written off** (for M1 it is dead — self-shielded, capped at 10^{-8}) **might be exactly where the E0 pairs are**. That is a genuinely different place to look.

7 Part III — How big is the E0 channel? (the open question)

Everything above — the channel structure, the selection rules, the opposite energy trends — is **settled**. The one thing still unknown is the *height* of the red curve: how strong the E0 (1S_0) capture is relative to the M1 (3S_1), i.e. $f = \sigma_{\text{E0}}/\sigma_{\text{M1}}$.

What the literature dig found

- The ${}^4\text{He}$ 0_2^+ (20.21 MeV) **E0 strength is real and measured**: its monopole transition form factor was measured precisely at MAMI [1], and is the subject of the “ α -particle monopole puzzle” [2]. So the *decay-side* E0 matrix element is established — it is a genuine *collective* monopole (“breathing”) transition, **not weak**.

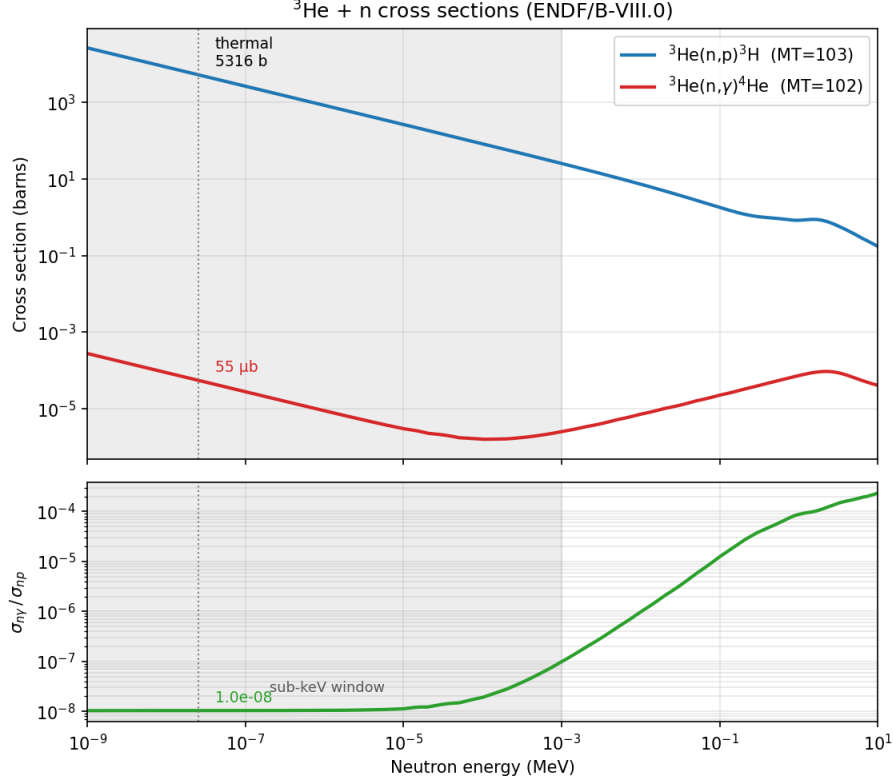


Figure 3: ${}^3\text{He}+n$ cross sections (ENDF/B-VIII.0). Bottom: the radiative/(n,p) ratio is flat at 10^{-8} through the sub-keV window and climbs to $\sim 2 \times 10^{-4}$ by 10 MeV — the E1 turning on. The E0 channel is not in this file.

- But that form factor is the *bound* $0_2^+ \rightarrow \text{g.s.}$ strength. Neutron capture needs the 1S_0 *continuum* $\rightarrow \text{g.s.}$ amplitude at 20.58 MeV (the 0_2^+ sits 0.37 MeV *below* threshold and enters through its sub-threshold tail). Converting one to the other is reaction theory, not a table lookup.
- The ATOMKI ${}^4\text{He}$ pair experiment (${}^3\text{H}(p, e^+e^-){}^4\text{He}$ [3]) models its pairs as the internal/external pair creation of *direct E1/M1 proton capture* — it does not isolate a clean E0 line, and the $p+t$ entrance couples to the 0_2^+ differently than $n+{}^3\text{He}$.
- I found **no calculation of ${}^3\text{He}(n, e^+e^-){}^4\text{He}$** (the E0 capture) anywhere — this specific channel appears genuinely un-computed.

A transparent bracket

Since no number exists, here is an honest order-of-magnitude argument. The decisive quantity is the ratio of *pair* yields, f/α_{IPC} :

- **Lower edge:** E0 is allowed and collective, and unpolarised $n+{}^3\text{He}$ populates the 1S_0 channel with statistical weight $\frac{1}{4}$ (vs $\frac{3}{4}$ for 3S_1). So f is unlikely to be far below the $\alpha_{\text{IPC}} \sim 10^{-3}$ scale.
- **Upper edge:** E0 pair emission has *no real-photon mode*, so it is intrinsically much slower than the M1 *photon* width — an E0 pair width is typically $\sim 10^{-2}$ – 10^{-3} of a comparable allowed M1 photon width. Even with the collective enhancement, $f \gtrsim 0.1$ is hard to reach.

$$\boxed{f = \sigma_{\text{E0}}/\sigma_{\text{M1}} \sim 10^{-3}\text{--}10^{-2}} \quad \Rightarrow \quad \frac{\text{E0 pairs}}{\text{M1 pairs}} = \frac{f}{\alpha_{\text{IPC}}} \sim 0.3\text{--}3 \quad (\text{up to } \sim 30)$$

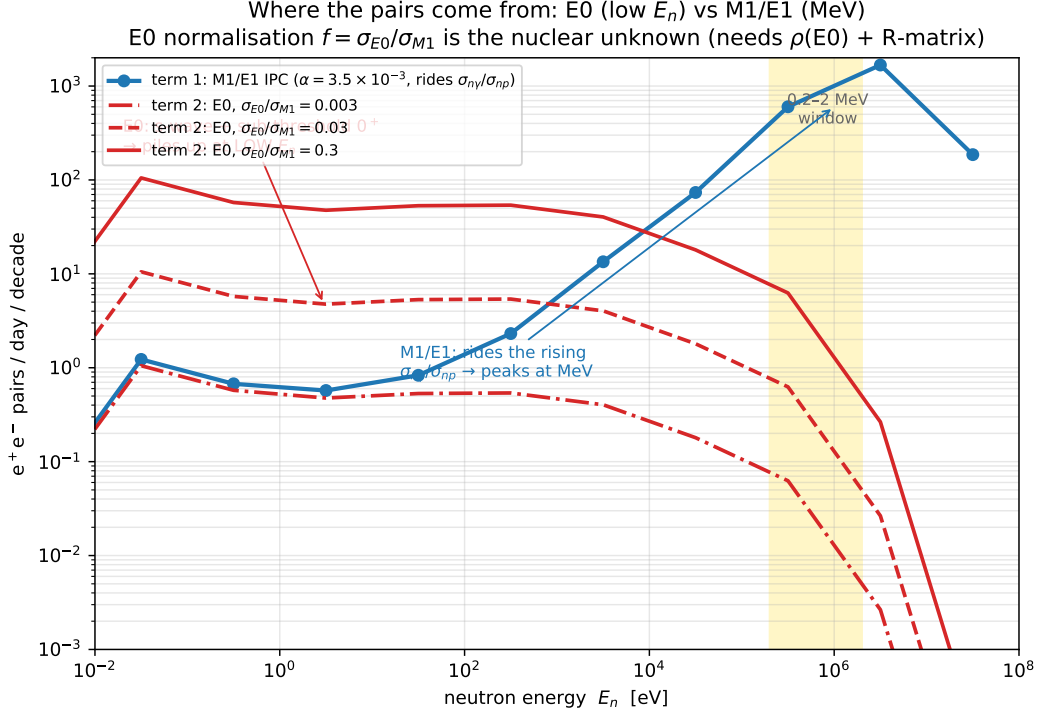


Figure 4: e^+e^- pairs/day vs neutron energy. M1/E1 (blue) peaks at MeV; E0 (red, three assumed strengths) piles up at low E_n . **Below ~ 1 keV the E0 channel can dominate the pair yield** even at a few-percent strength, because it dodges the 0.35% IPC suppression. The vertical position of the red curves is the one unknown (Part III).

at the low neutron energies where E0 lives. **So E0 is plausibly a same-order-or-larger contributor to the sub-keV pair yield** — which is exactly the “same order of magnitude” point that started this. The three red curves in Fig. 4 ($f = 0.003/0.03/0.3$) bracket this range.

What would replace the bracket with a number

One calculation: a ${}^4\text{He}$ **multi-level R-matrix** (G. M. Hale maintains the evaluated ${}^4\text{He}$ system) for the 1S_0 $n + {}^3\text{He} \rightarrow \text{g.s. E0 capture}$, *normalised* by the measured (e,e') monopole strength and the 0_2^+ neutron asymptotic normalisation. That is a concrete, well-posed ask for a ${}^4\text{He}$ -theory contact, and is the clean next step if we want to firm up the red curve. (*Worth keeping on the list to work on together.*)

Reproducing the figures

Schematic `make_capture_breakdown_fig.py` (Fig. 1); doorway calc `calc_doorway_formation.py` (Fig. 2); cross sections `plot_he3_xs.py` (Fig. 3); pair yield `make_e0_pair_yield_fig.py` (Fig. 4). Companion markdown: `docs/e0_branch/` (`capture_breakdown.md`, `doorway_states_note.md`, `ipc_estimation_method.md`, `ipc_roadmap.md`).

References

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